

The Pent-athlon

We give the Athlon 64 FX a run for its hype

Processors built on the 64-bit architecture are slowly, but surely sneaking into the desktop environment. SUN Microsystems, IBM, Digital and HP have been using the architecture for many years, but all of these were tied down to server-class operating systems that were specifically written to make use of their special instruction sets. Moreover, all these were pure 64-bit architectures that were incapable of running existing 32-bit applications. The AMD Athlon 64 FX makes all this history. This AMD processor is a desktop version from the Opteron family. We pitched it against the tried and trusted Pentium 4 3.2 GHz that is currently ruling the high-end desktop market.

This Athlon differs from its predecessor, the Barton core Athlon XP 3200+. Apart from two additional integer pipelines, it has an on-chip memory controller and HyperTransport controller making the I/O system faster. It operates in three modes:

Long mode: In this mode, the operating system performs in pure 64-bit mode, and supports only 64-bit applications.

Compatibility mode: Here, it supports both 32 and 64-bit applications under a 64-bit operating system.

Legacy mode: The legacy mode can only run current 32-bit operating systems; 64-bit applications cannot be used.

We tested the Athlon 64 in Legacy mode, running a 32-bit version of Windows XP as it will have to compete with a 32-bit Intel family, for at least the next year. The results were compared with those given by a Pentium 4 3.2 GHz on a Canterwood (Intel 875 PBZ) motherboard. Both the computers were configured with 512 MB RAM and used the Gainward GeForce FX 5900 Ultra graphics card. The overall analysis was a combination of real-world and synthetic scores.

The the real world test, application benchmarks included Photoshop tests, audio and video encoding and games that

involved intense calculations such as calculating the value of pi to varying degrees of precision, ranging from 16K to 32 million decimal places. In the synthetic benchmarks, we used Sisoft Sandra 2003 Pro, Winstone Content Creation 2003 and PCMark 2002 Pro.

Pure performance

While the time difference between the pi calculations was small at 1 million decimal places (6 secs), it increased drastically as the decimal places went up to 32 million (409 secs).

In SiSoft, the Dhrystone and Whetstone scores were much lower than those of the Athlon XP 3200+. This seemed very unlikely, so we disregarded them. We installed Red Hat 9 on both the computers, and configured them in exactly the same way—including switching off GUI and networking—and stopped all services that were not needed. We then compiled the unixbench v 4.1 benchmark to generate code that works well with their respective architectures. The results clearly showed that the Athlon FX has an upper hand in both, integer and floating point operations, but it also highlighted an impor-

tant problem. Many applications may have to be specifically recompiled for the Athlon, since they may not be optimised for its specific enhancements.

Gaming scenes

In the gaming tests, the Athlon FX gave quite disappointing results. In *Quake III*, it scored better than the Athlon XP 3200+, but when compared with the P4, it lagged behind by almost 100 fps. In *Gunmetal*, the P4 scored better at all resolutions with low-quality settings. However, when the settings were turned to high, the Athlon zoomed ahead.

Office suitability

In the Winstone Content creation tests, the Athlon got 62.3 Winstones in the multimedia content creation, and the P4 logged 43.3.

In the business content creation, it logged 41.9, while P4 scored 35.3. The WinRAR compression, the Athlon is almost twice as fast as the P4, but do you really need a 64-bit computer to run office applications? In the PCMark 2002 CPU tests, the scores again plummeted, with the P4 scoring 7939 points to the Athlon's 7100. This again may be because of the lack of optimisation, but it also emphasises that finding applica-

